Why do the NHS1 forecasts look weak?

Diagnostic audit | 23 October 2026 | original benchmark preserved

There is genuine underfitting, particularly in the saved native AutoGP fits. The requested RHS is substantially closer to the DGP oracle, and most of its conditional expected squared error is irreducible noise. Its remaining gap is meaningful. The evidence points to compressed/correlated reservoir features and limited kernel-space exploration, rather than an all-zero RHS coefficient vector or a general need for deeper models.

Model / diagnostic	Pooled RMSE	CRPS	NLL
DGP oracle	0.51685	0.28552	0.73227
AutoDESN RHS: 0.001 / 1	0.57887	0.32388	0.88675
AutoDESN ridge	0.63934	0.35865	0.97752
Native AutoGP	0.66644	0.38656	1.10323
AR90_phase60	0.54966	0.30653	0.82165
Quasiperiodic GP diagnostic	0.58178	0.32491	0.88021

The oracle knows the generating mechanism. The new AR and GP rows are post-hoc diagnostic fits, selected/fitted on training data only; they are not replacement Bayesian AutoDESN/AutoGP results or a new predeclared superiority benchmark. The original four model score tables remain byte-identical. All full-grid rows use 471 origins and 30 horizons (14,130 dependent pairs).

Main conclusions

Forecasting noise matters: oracle pooled RMSE is 0.51685 versus requested RHS 0.57887. Missing future spikes is not by itself evidence of a bad conditional-mean forecast.

RHS has not shut down: a typical draw has about 20 standardized slopes above 0.1 in magnitude; exact zeros are absent.

Representation matters: direct lag features, with a training-selected seasonal-phase reference, improve pooled RMSE to 0.54966.

Native AutoGP missed a high-density in-grammar configuration. A depth-three diagnostic reaches RMSE 0.58178 and has a much larger native log joint than saved equal-dimensional particles.

Variance modeling is a separate problem: a constant observation variance cannot reproduce the DGP volatility clusters.

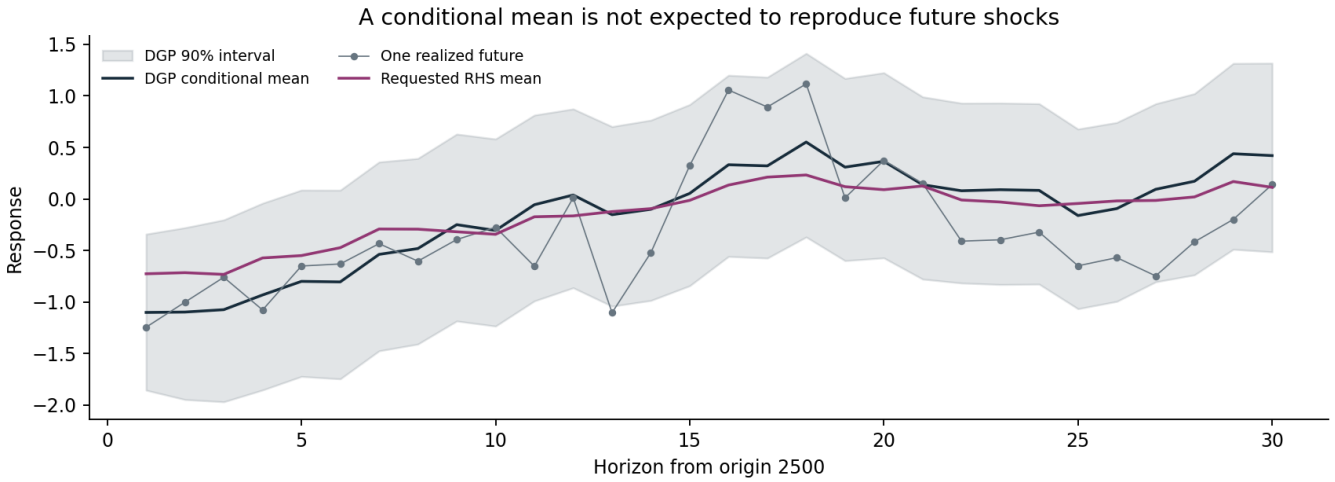
1. What is genuinely unpredictable?

For information F(o) available at origin o, let m*(o,h)=E[Y(o+h)|F(o)] and V*(o,h)=Var[Y(o+h)|F(o)]. For any point forecast a(o,h), the conditional squared risk is V*(o,h)+(a(o,h)-m*(o,h))^2. The first term is irreducible even with the exact DGP. The second is an avoidable conditional-mean error.

The audit simulates 32,768 independent futures at each of the 471 origins, recursively evolving the nonlinear residual and GJR variance. The deterministic season is known to the oracle. Only observed history initializes a forecast; future observations enter scoring, not simulation. An independent R continuation agrees to 5.1e-15. The current variance is recoverable from observed history and the known DGP to 2.0e-15.

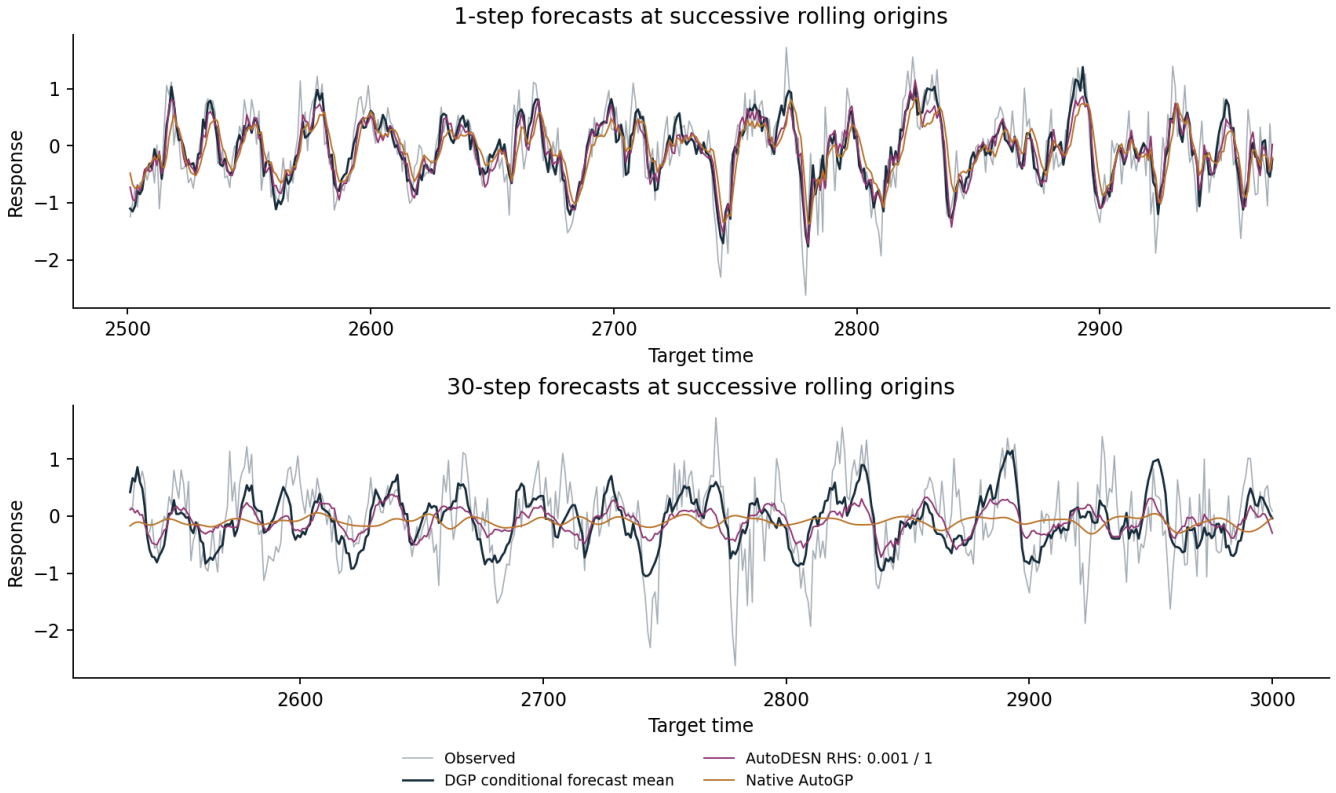
Requested RHS risk	Horizon 1	Horizon 30	Pooled
Actual model RMSE	0.4462	0.5965	0.5789
Irreducible conditional SD	0.4205	0.5763	0.5437
RMSE to oracle conditional mean	0.1786	0.2977	0.2884
Irreducible fraction of expected MSE	0.8472	0.7894	0.7803

These fractions concern the average of per-origin conditional risks on the realized histories, not a decomposition of the one realized squared-error sample. The oracle realized RMSE (0.51685) differs from the square root of average conditional variance (0.54366), because the held-out shocks are one dependent realization. No iid standard error over rolling pairs is used.

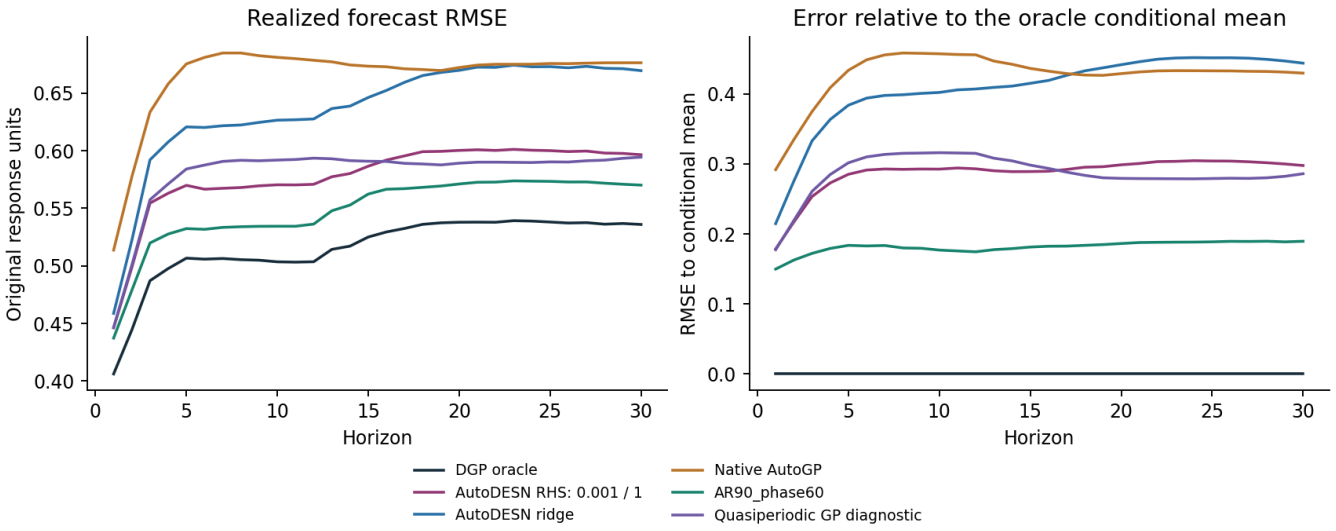


Oracle conditional-mean Monte Carlo SE: RMS 0.00180, maximum 0.00277. Half-budget versus full-budget pooled RMSE is 0.516849 versus 0.516849. These computational replications quantify simulation error, not variation across independent time series.

2. A flat-looking forecast can still be informative



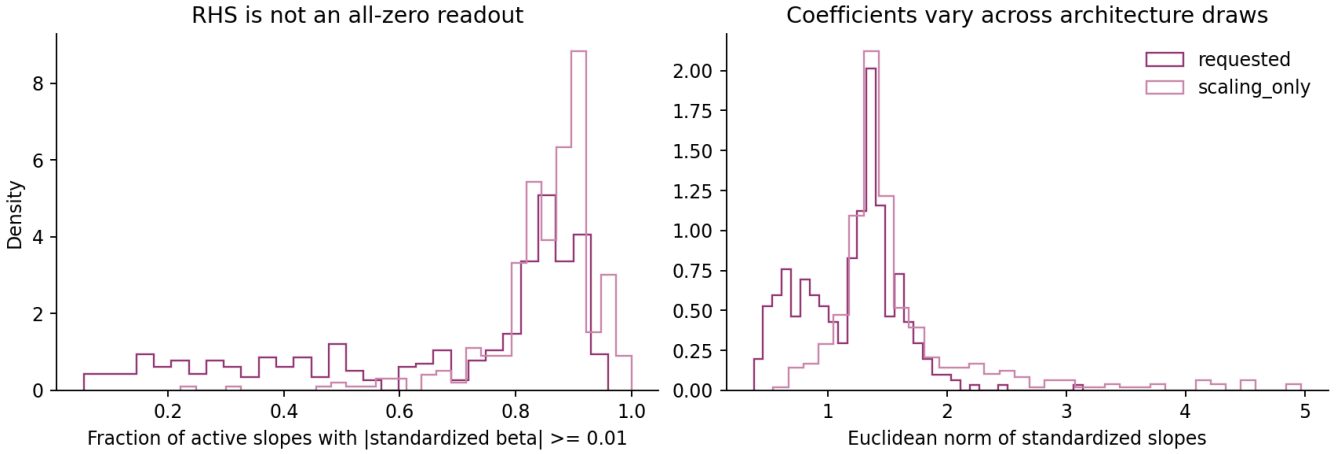
The earlier horizon-30 plot joined 471 different forecasts, each issued 30 observations before its target. It did not show one-step fitted values or a single 471-step forecast path. At horizon 30 the conditional mean must average over many unknown intervening shocks. The correct benchmark is the simulated oracle conditional mean, not the future realized one-step DGP mean.



3. Are the RHS coefficients mostly zero?

No. This normalized RHS is continuous shrinkage, not a prior with an atom at zero. Coefficient magnitudes are inspected in the frozen standardized readout basis; raw-coordinate coefficients would give a misleading comparison because their units differ. Coefficients from different reservoirs are not treated as the same variable.

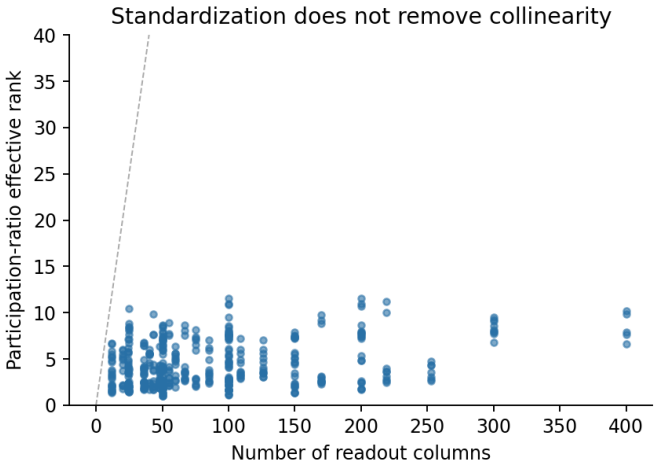
Quantity (median over draws)	Requested .001/1	Control 1/2
Active slope dimension	50	50
Fraction with |beta| < 0.01	0.1882	0.12
Number with |beta| > 0.1	19.5	22
Slope Euclidean norm	1.282	1.408
Exact-zero count	0	0

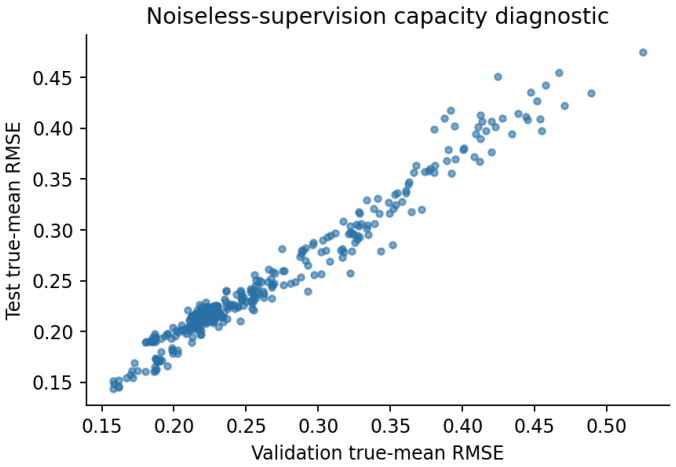


Conditional on leading architecture 118, the requested fit has 20 of 50 slopes with conditional 95% intervals excluding zero (121 compact draws for that architecture). This is not a variable-inclusion probability, and correlated coordinates make individual coefficient interpretations fragile. The leading geometry has participation-ratio effective rank 7.15 among 50 standardized columns. Its full numerical rank is 50; these are different notions.

The learned global scale tau is not fixed at tau0: requested posterior median tau=0.07463 although tau0=0.001. The control median is 0.1404. Slab-variance medians are 0.7817 and 3.537. Relaxing these prior controls changes shrinkage, but only modestly changes the original forecast scores. This does not support wholesale removal of shrinkage as the first remedy.

4. What the finite reservoir bank can represent





Each of the 426 frozen feature sets was fitted to two targets: ordinary noisy responses, and the privileged true one-step conditional mean. Ridge penalties and candidates were selected on rows 501:2000 versus validation rows 2001:2500, then refitted on 501:2500. Test rows 2501:3000 are 500 one-step observations here, distinct from the original 471-origin horizon-1 subset. The true-mean target is a representational diagnostic, unavailable in a real application.

Mean-function diagnostic (500 test rows)	True-mean RMSE
Original requested posterior, 384 draws	0.1812
Noiseless supervision, leading model 118	0.1714
Noiseless supervision, validation-selected bank model 189	0.1514
Leading 118 plus explicit phase, noiseless supervision	0.1138
Direct lags + phase, noiseless supervision	0.0799
Direct lags + phase, ordinary noisy training	0.1231

The best true-mean validation model has two 50-unit layers and 90 input lags. Its fitted slope norm is about 39 and condition number 8251. Weak, correlated feature directions can require large coefficient combinations; a larger count of neurons is not equivalent to a useful or well-conditioned basis. The earlier diagonal shrinkage proxy near 44 is not a true degrees-of-freedom count and not a count of selected variables.

The original requested mean-function error is similar in training and test (0.1854 versus 0.1812 for 384 draws). This is not the classic pattern of excellent training fit followed by failed generalization. Finite representation, regularization and estimation all contribute; these diagnostics do not uniquely separate them.

5. Controlled direct-lag and calendar diagnostics

Six linear ridge families were fit: lags {1,2,12,60}, all lags 1:60 and all lags 1:90, each with or without 59 phase indicators for period 60. The period and four-lag structure are DGP-informed; the 60/90 lag windows are generic. Each penalty was selected by rolling 30-step validation entirely within the original training data, and all parameters were then frozen at 2500. The global validation winner was AR90 plus phase.

Family	Validation RMSE	Test RMSE	CRPS
AR4	0.6172	0.5619	0.3111
AR60	0.6210	0.5619	0.3120
AR90	0.6192	0.5558	0.3082
AR4_phase60	0.5871	0.5386	0.3008
AR60_phase60	0.5932	0.5469	0.3042
AR90_phase60	0.5857	0.5497	0.3065

The lower test score of the four-lag/phase case is reported transparently; it was not the validation-selected winner. Choosing it after viewing the test would be another selection step, not evidence of a prospectively superior model. These plug-in Gaussian references are not Bayesian architecture averages.

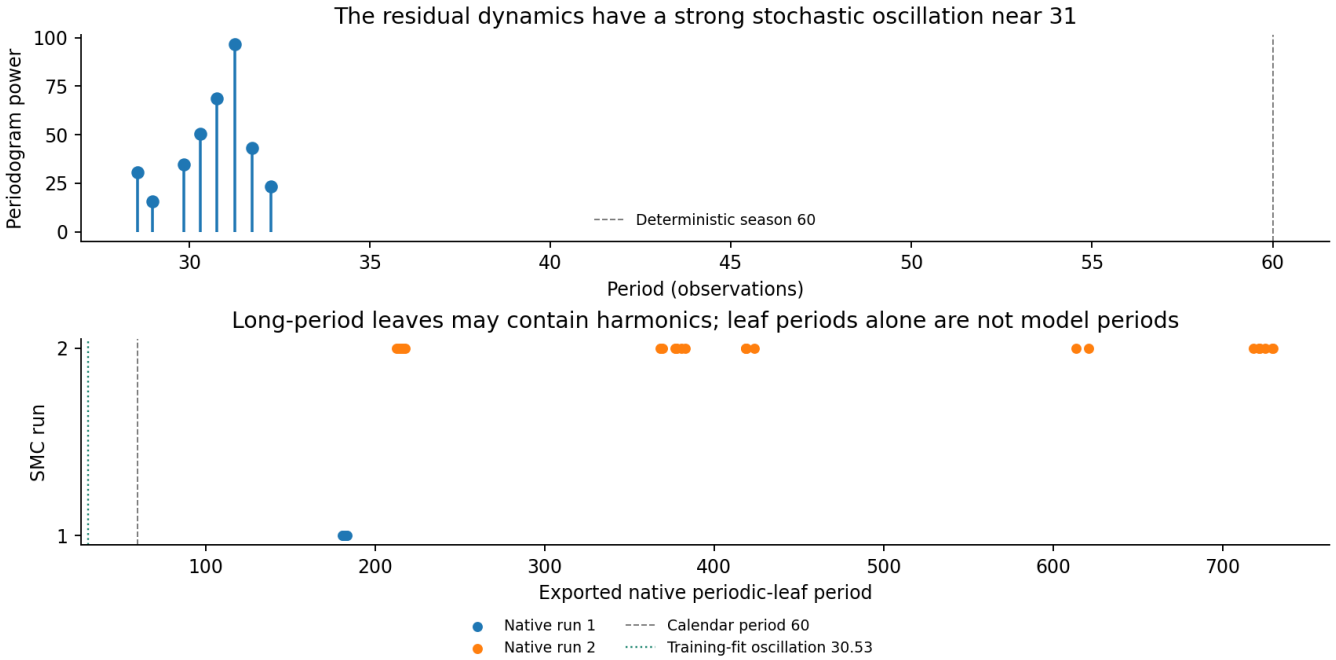
What this establishes

A simple direct-lag representation can recover signal that the current randomized reservoir/readout combination misses. AR90 without phase also improves on requested RHS, so the evidence is not solely a benefit from being told the seasonal period. Adding a direct-lag skip connection is a more targeted next experiment than indiscriminately increasing network depth.

The phase-only climatology performs poorly (pooled RMSE 0.7105), as do seasonal-naive 60 (0.7070), persistence (0.9696), and the training-mean reference (0.6786). The training and test empirical phase profiles correlate only 0.358. Residual dynamics have strong stochastic oscillations near 31 observations, so this is not just a fixed seasonal curve plus independent noise.

The validation-selected lag/phase reference still has only 87.9% overall coverage for its nominal 90% interval. Better point forecasts do not automatically solve the heteroscedastic calibration problem.

6. Native AutoGP: capacity versus search



Native run	Particle ESS	Distinct tree structures	Effective number of structures
1	5.46	2	1.02
2	4.24	4	2.58

Different continuous parameter values had previously been counted as distinct kernel states. Their number is not structural diversity. Commutative and associative kernel products/sums are canonicalized. The effective structure count is inverse-Simpson category diversity, not a Monte Carlo ESS estimate. Particle weight ESS describes the current particle set, not whether it has found all relevant kernel regions. Long-period leaves may express shorter harmonics, so absence of an explicit 60- or 31-period leaf alone is not proof of incapacity.

Native time is scaled to[0,1] across 1999 observation intervals. Under its log-normal period prior, P(period<=60)=0.0224, and P(54<=period<=66)=0.0107. Short cycles occupy a relatively small part of the default prior. Only eight particles per run and shallow local rejuvenation were used. These are plausible search limitations, not a demonstrated error in the native target density.

7. A stronger GP already fits inside the grammar

A depth-two known-period diagnostic used Periodic 60 + OU. A second depth-three diagnostic used Periodic 60 + Periodic(P)*GammaExponential, initialized by the training periodogram at 31.25 and fitted to P=30.527 with coherence length 33.43. Both fit parameters by training marginal likelihood with two starting points; neither retunes on the held-out scores. They are plug-in diagnostics, not rerun Bayesian SMC fits.

Fit	RMSE	CRPS	NLL
Native AutoGP	0.66644	0.38656	1.10323
Fixed-period GP diagnostic	0.67314	0.37785	1.01825
Quasiperiodic GP diagnostic	0.58178	0.32491	0.88021
AutoDESN RHS: 0.001 / 1	0.57887	0.32388	0.88675

Known period 60 by itself improves distributional calibration but does not improve pooled point error. The additional training-identified stochastic oscillation brings GP performance close to RHS. Its depth 3 is below the original maximum 4, so a deeper grammar is not required to contain this configuration.

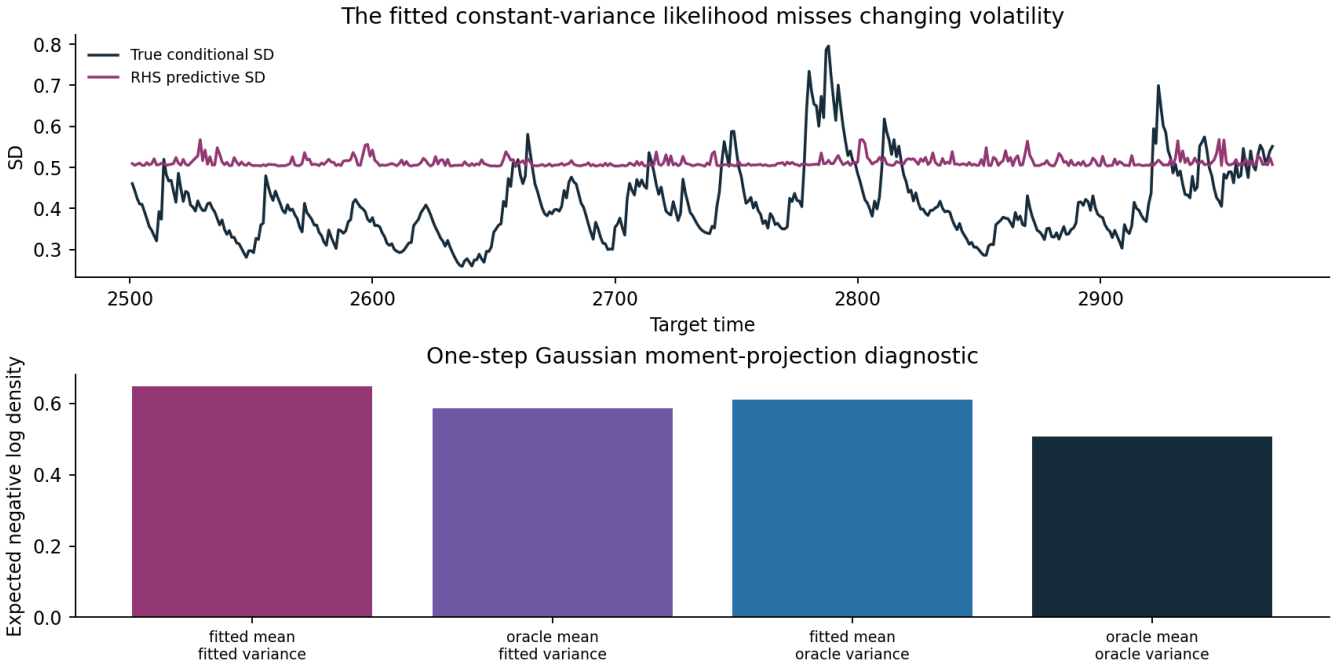
Native prior-adjusted check

Configuration	Latent dimension	Native log joint
Best saved particle with matching dimension	10	1337.71
Training-fitted depth-three candidate	10	1788.42
Difference		450.72

The Julia Gen calculation uses the original AutoGP priors, tree probabilities, parameter coordinates, data scaling and likelihood normalizers. This is strong pointwise evidence of a missed high-density region. It is not a Bayes factor, an integrated posterior architecture probability, or proof that the diagnostic has a converged Bayesian posterior. A better SMC exploration audit is required before treating the original native GP result as representative of its model class.

Independent reconstruction of every native kernel reproduces the exported initial-origin means to 4.43e-14. The evidence points to inference/model representation, not a discovered forecast-export or conditioning bug. With fixed kernels and weights, a time-indexed GP conditional mean is affine in subsequently observed responses. A GP on lagged responses would provide a different nonlinear autoregressive input model.

8. Mean flexibility and conditional variance are different



Conditional mean	Predictive variance	Expected NLL	Expected CRPS
fitted	fitted	0.64750	0.25663
oracle	fitted	0.58663	0.23761
fitted	oracle	0.61123	0.25376
oracle	oracle	0.50743	0.23175

This 2x2 calculation projects the fitted one-step distribution to its Gaussian mean/variance and integrates over the exact Gaussian DGP conditional response. It is not the original posterior-mixture score. It isolates both mean error and variance error; their improvements need not add because there is an interaction.

Requested RHS nominal 90% one-step coverage is 96.8% in low-volatility rows but 83.0% in high-volatility rows. Its average observation variance is about 0.2508, while mean true conditional variance is 0.2068 in training and 0.1768 over the horizon-1 evaluation rows. Mean approximation and posterior uncertainty can be absorbed into a constant residual variance; the test block also happens to be calmer. Adding mean parameters alone does not specify a conditional variance process.

9. What to change next, and what not to conclude

First: preserve useful predictors

Add direct lag features alongside reservoir states, and a separate explicit calendar/seasonal component when the period is known or selected inside training. Test these as controlled ablations before expanding every reservoir. Use group-aware prior scales so direct lags, calendar terms and reservoir features have deliberate regularization.

Second: repair the AutoGP exploration comparison

Use a broader scientifically justified period prior or training-informed proposals with the exact correction, more particles and independent runs, and seed diagnostics near both discovered period modes. Monitor tree diversity, conditional periods, log-joint levels and predictive functions. Increasing depth alone does not fix missed regions. Any changed prior is a new model comparison, not the same posterior.

Third: model conditional volatility

Introduce a validated variance component with its own priors and causal forecast recursion. A residual-based GARCH variance can depend on the mean coefficients, so Gaussian coefficient-collapsed moves cannot simply be reused without checking the new conditionals. Keep the first mean-representation comparison separate from the later variance-model extension.

Keep regularization, but audit its geometry

The present evidence does not justify setting shrinkage to zero. Compare group scales, direct-lag skip features and optionally orthogonalized bases under explicitly defined priors. Whitening changes an independent-coordinate RHS prior; it is not a numerically neutral transformation.

Reassess on independent realizations

The oracle and diagnostic models were studied post hoc on one selected series. Period 60 is privileged information for the references. Training-only tuning prevents direct test selection of their parameters, but does not turn this audit into a fresh predeclared benchmark. Confirm improvements on new seeds and periods; overlapping rolling forecasts and Monte Carlo paths are not independent DGP replications.

10. Checks, reproduction and sources

All original four score tables were compared byte-for-byte against the committed numeric archive. The new feature-based one-step reconstruction matches original AutoDESN forecasts to 5.11e-15. Native GP covariance-kernel reconstruction matches exported means to 4.43e-14. Independent R/Python DGP continuations match to 5.11e-15. Gaussian triangular prequential conditioning and the future-data perturbation check pass. The depth-three likelihood gradient agrees with finite differences to 3.07e-6.

The source scripts and tables distinguish: the unchanged Bayesian benchmark; a privileged DGP oracle; noiseless mean-supervision capacity diagnostics; training-selected plug-in lag references; and training-likelihood GP configurations. No diagnostic is silently substituted for the original posterior average. Gaussian smoothing at training locations is kept separate from causal one-step fitted means, which do not condition on future training responses once the fitted parameters are held fixed. Parameters were estimated on the full training block; these are in-sample diagnostics, not past-only refits.

Run from the repository root: audit/oracle.py; audit/diagnostics.R; audit/lag_ablation.py; audit/gp_audit.py; audit/gp_quasiperiodic.py; audit/native_score.jl; audit/test_oracle.R; audit/test_audit.py; audit/analyze.py; audit/report.py. See audit/README.md for full paths, the executed environment and the order needed for the native score check. Original tests passing is not a proof of posterior convergence, and this audit does not certify a new Bayesian AutoGP fit.

Research context

Piironen, J. and Vehtari, A. (2017). Sparsity information and regularization in the horseshoe and other shrinkage priors. Electronic Journal of Statistics. https://arxiv.org/abs/1707.01694. The regularized horseshoe is continuous shrinkage with a finite slab, not exact spike-and-slab selection.

Grigoryeva, L. and Ortega, J.-P. (2018). Echo state networks are universal. Neural Networks 108:495-508. https://arxiv.org/abs/1806.00797. This is a class-level approximation theorem with stated assumptions, not a guarantee for a particular finite random reservoir. For stochastic-input Lp extensions, see Gonon and Ortega, https://arxiv.org/abs/1807.02621.

Saad et al. (2023). Sequential Monte Carlo Learning for Time Series Structure Discovery. ICML/PMLR 202. Native implementation: https://github.com/probsys/AutoGP.jl, audited pinned revision 2ad372db2b8bea176e3b32a769bfb5d6cff80866, AutoGP 0.1.19, Julia 1.10.10. Exact code inspection and native Gen evaluations, rather than generic GP claims, support the diagnostics here.

The inherited package release-check portability/documentation issues remain documented in the original report. No new package-wide release check or new production model is claimed by this diagnostic audit.